StableHGB: Stable Index Timing with Histogram Gradient Boosting

Yi Ji* **Haozhou Lang*** **Letian Liang*** **Haoxiang Rui***
School of Future Technology
South China University of Technology
*Equal contribution. Authors are listed in alphabetical order by surname.

Abstract

Financial market timing is challenging because predictive signals are weak, daily labeled samples are scarce, and the calibration of model probabilities can shift across market regimes. We propose StableHGB, a machine-learning-based timing framework for a micro-cap equity index. The framework trains a histogram gradient boosting classifier to predict five-trading-day price direction from technical features, and couples the classifier with a policy layer that converts probability estimates into binary market exposure. The policy layer combines two components: Relative Signal Stabilizer, which replaces fixed probability thresholds with prior-only expanding historical ranks and confirmation-based switching, and Trend Position Guard, which restores full exposure when moving-average alignment indicates a favorable trend regime. Policy parameters are selected using non-overlapping rolling-validation windows from 2022 to 2024, while the final classifier is trained only on labels whose target windows end before 2024. On the held-out investment period from January 2, 2025 to May 6, 2026, under close-to-close accounting and zero transaction cost, StableHGB achieves a cumulative return of 153.44%, a Sharpe ratio of 3.38, and a maximum drawdown of -9.84%, compared with 109.21%, 2.30, and -17.26% for buy-and-hold. Under a simplified 20-basis-point one-way turnover cost, cumulative return remains 147.46%. The classifier attains a ROC-AUC of 0.635, while fixed-threshold accuracy remains below the majority-class baseline, indicating that performance is driven primarily by probability ranking and validation-selected position construction rather than direct classification accuracy. These results provide historical evidence that a rank-based policy layer can improve the usability of noisy timing signals, while leaving cross-instrument and cross-regime generalization as open limitations. Code is available at `https://github.com/V1nc3ntL1ang/StableHGB`.

1 Introduction

Market timing seeks to adjust market exposure according to expected future conditions rather than remaining continuously invested. The task is especially challenging for a volatile micro-cap index: daily observations have a low signal-to-noise ratio, nonlinear interactions may exist among technical indicators, and a strategy must convert imperfect predictions into positions without creating excessive turnover. A model can also preserve useful ranking ability while its absolute probability level changes across periods, making a fixed threshold such as 0.5 an unstable trading rule.

Classical technical strategies use moving averages, momentum, and volatility filters to construct interpretable timing signals. Machine-learning methods can combine a larger set of indicators and capture nonlinear interactions, but their predictions do not by themselves define an executable portfolio policy. In particular, strong classification accuracy is neither necessary nor sufficient for a successful timing strategy: what matters is how predictions are ranked, validated, translated into exposure, and evaluated under a leakage-controlled historical protocol.

This paper presents StableHGB, a machine-learning timing framework for the SH#880823 micro-cap index. The prediction engine is a strongly regularized `HistGradientBoostingClassifier`. Its estimated positive-class probabilities are processed by two proposed policy components. The Relative Signal Stabilizer expresses each probability relative to earlier model outputs and changes the position only after consecutive entry or exit confirmations. The Trend Position Guard restores full exposure whenever the moving averages satisfy a bullish alignment condition. The final policy therefore switches between zero and full exposure rather than using continuous position sizing.

The main contributions are as follows:

- We formulate micro-cap index timing as five-trading-day direction prediction followed by an explicitly validated probability-to-position policy.
- We introduce the Relative Signal Stabilizer, which combines a prior-only expanding probability rank, a two-day confirmation mechanism, and an entry–exit deadband to reduce dependence on the absolute probability level.
- We introduce the Trend Position Guard, which preserves full exposure during bullish moving-average alignment, and evaluate both components through rolling validation, component ablations, and transaction-cost sensitivity analysis.

The reported results show higher terminal wealth and a higher test-period Sharpe ratio than the evaluated traditional and machine-learning baselines. The analysis also documents important qualifications: the fixed-threshold classifier accuracy is below the majority-class baseline, the 2024 validation fold has a maximum drawdown of -40.84%, and the backtest uses a simplified close-to-close execution convention. The evidence should therefore be interpreted as a historical evaluation of a policy-design pipeline rather than as a universal profitability claim.

2 Related Work

2.1 Technical trading and time-series momentum

Moving-average and momentum rules are among the most widely studied forms of technical trading. Brock et al. evaluate simple moving-average and trading-range rules on long-horizon equity data and show that technical rules can contain information not summarized by standard return models [3]. Lo et al. develop computational representations of technical patterns and examine their statistical content [9]. Time-series momentum studies further document that an asset's own past return can be associated with its future return across multiple markets [10]. These findings motivate the trend, momentum, volatility, and rebound features used in this project, while also underscoring the need for strict out-of-sample evaluation.

2.2 Machine learning for financial prediction

Tree ensembles provide flexible nonlinear predictors without requiring feature standardization. Random forests aggregate randomized decision trees [2], while gradient boosting constructs an additive model by fitting successive learners to negative loss gradients [5]. Histogram-based implementations reduce split-search cost by operating on binned feature values, and LightGBM extends this design for efficient large-scale learning [7]. Machine-learning methods have also been applied to return prediction and empirical asset pricing, including support-vector machines, recurrent neural networks, and nonlinear cross-sectional models [4, 6, 8]. The present work differs from studies focused only on predictive accuracy: the central object is the complete mapping from a probability forecast to an executable exposure policy.

2.3 Backtest selection and data-snooping risk

Evaluating many models or policies on the same historical period can produce optimistic results through data snooping [13]. This risk is particularly relevant when position thresholds are selected from a large grid. StableHGB therefore separates chronological training, rolling policy validation, labeled predictive testing, and the final investment backtest. The protocol does not eliminate selection risk, but it avoids using the held-out 2025–2026 period to choose the reported policy parameters and explicitly reports validation-fold behavior.

3 Method

3.1 Problem formulation

Let C_t denote the index close on trading observation t, and let $\mathbf{x}_t \in \mathbb{R}^{20}$ denote the feature vector available at that observation. The predictive target is the five-trading-observation direction

$$y_t = \mathbb{I}(C_{t+5} > C_t)\,, \tag{1}$$

where $\mathbb{I}(\cdot)$ is the indicator function. A classifier f_θ produces an estimated positive-class probability

$$p_t = f_\theta(\mathbf{x}_t) \in [0, 1]. \tag{2}$$

The trading problem is then to map the sequence $\{p_t\}$ and contemporaneously available policy features into a target exposure $z_t \in \{0, 1\}$.

3.2 Data and feature construction

The raw dataset contains 2,246 daily OHLCV records for the SH#880823 micro-cap index from February 3, 2017 to May 6, 2026. The data were exported from Tongdaxin and are marked as unadjusted. Rolling-window construction and removal of incomplete feature rows yield a feature-complete sample beginning on June 1, 2017. Infinite feature values are replaced with missing values, and rows lacking required model features are removed.

The model uses the 20 features summarized in Table 1. The set covers trend level, trend slope, momentum, volatility, rebound, MACD, Bollinger-band information [1], and volume abnormality. The 20-day moving average is computed as an intermediate quantity but is not included directly in the model vector. It is used to construct the 20-day slope, Bollinger bands, the MA20 baseline, and the policy-only alignment indicator.

Table 1: Model input features. The policy-only `ma_alignment` variable is excluded from the classifier.

Category	Features	Construction
Moving averages	`ma5`, `ma10`, `ma60`	Simple moving averages of closing price
MA slopes	`ma20_slope`, `ma60_slope`	$\mathrm{MA}_N(t)/\mathrm{MA}_N(t-5) - 1$
Price deviation	`close_vs_ma60`	$C_t/\mathrm{MA}_{60}(t) - 1$
Momentum	`momentum20`	$C_t/C_{t-20} - 1$
Volatility	`volatility5`, `volatility20`	Rolling standard deviation of daily returns
Volatility rank	`volatility20_rank`	Rolling percentile rank over up to 252 observations
Trend strength	`trend_strength20`, `trend_strength60`	MA slope divided by 20-day volatility
Rebound	`rebound_from_20d_low`, `rebound_from_60d_low`	Close relative to the rolling minimum
MACD	`macd_signal`, `macd_histogram`	EMA(12)–EMA(26) signal and histogram
Bollinger bands	`bb_upper`, `bb_lower`, `bb_width`	20-day mean ± 2 standard deviations and relative width
Volume	`volume_zscore20`	20-day volume z-score

The policy-only trend feature is

$$g_t = \mathbb{I}(\mathrm{MA}_5(t) > \mathrm{MA}_{20}(t) > \mathrm{MA}_{60}(t))\,. \tag{3}$$

This feature is used exclusively by the Trend Position Guard and is not supplied to the classifier, thereby separating predictive information from the deterministic policy override.

3.3 Histogram gradient boosting classifier

Histogram gradient boosting constructs an additive decision-tree model. At boosting iteration m, a new tree h_m is fitted to the negative gradient of the binary classification loss under the current model F_{m-1}, and the ensemble is updated as

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta h_m(\mathbf{x}), \tag{4}$$

where η is the learning rate. Rather than evaluating every raw feature value as a possible split, the implementation maps each continuous feature into at most 255 integer-valued bins and searches over bin boundaries. This reduces split-search and memory costs without implying that the bins are equally spaced.

The final classifier uses the scikit-learn implementation [11] with the fixed settings summarized in Table 2. These model settings are not part of the 108-candidate policy search. The small learning rate, limited number of leaves, and large minimum leaf size are intended to constrain model complexity in the small-sample setting. Because tree splits depend on order and thresholds rather than Euclidean feature scale, no standardization is required for this classifier.

3.4 Relative Signal Stabilizer

A fixed probability threshold can be sensitive to shifts in the absolute level of p_t. The Relative Signal Stabilizer instead computes a prior-only expanding rank. For $t > 1$,

$$r_t = \frac{1}{t-1} \sum_{j<t} \mathbb{I}(p_j \le p_t), \qquad r_1 = 0.5. \tag{5}$$

Ties are counted on the right side of the empirical distribution, matching the implementation. The transformation does not guarantee a uniform rank sequence; it expresses the current prediction relative to earlier model outputs and thereby reduces dependence on the probability scale.

The selected policy uses an entry threshold $u = 0.55$, an exit threshold $\ell = 0.45$, and two consecutive confirmations. A target exposure of one is entered only after $r_t \ge u$ for two consecutive observations, while exposure is changed to zero only after $r_t \le \ell$ for two consecutive observations. When $\ell < r_t < u$, or when a threshold has not yet received the required confirmations, the previous target exposure is retained. The interval $(0.45, 0.55)$ therefore forms an entry–exit deadband. The selected smoothing window is one, so no additional moving-average smoothing is applied.

3.5 Trend Position Guard

Let $s_t \in \{0, 1\}$ denote the target exposure produced by the Relative Signal Stabilizer. The Trend Position Guard produces the final target exposure

$$z_t = \begin{cases} 1, & \text{if } g_t = 1, \\ s_t, & \text{if } g_t = 0. \end{cases} \tag{6}$$

Thus, bullish moving-average alignment overrides a zero timing position and restores full exposure. The component is designed to preserve participation during aligned upward trends; it does not guarantee lower drawdown, and the ablation study shows that the guard can increase drawdown when alignment is followed by reversal.

The fixed classifier settings and the validation-selected position-policy settings are summarized jointly in Table 2. Grouping them in one table distinguishes parameters fixed before policy search from parameters selected by rolling validation.

Table 2: Fixed classifier configuration and validation-selected position-policy configuration.

Component	Setting	Value
Classifier	Boosting iterations	150
	Learning rate	0.05
	Maximum leaves per tree	15
	Minimum observations per leaf	70
	Random seed	42
Relative Signal Stabilizer	Entry rank	0.55
	Exit rank	0.45
	Consecutive confirmations	2
	Exposure range	[0, 1]
	Smoothing window	1
Trend Position Guard	Alignment trigger	$g_t > 0.5$
	Minimum guarded exposure	1.0

3.6 Rolling policy selection

Policy parameters are selected using three non-overlapping annual validation windows: 2022, 2023, and 2024. For each fold, the classifier is fitted using labels whose five-trading-day target windows end before the fold begins. Validation labels must both start and end within the fold, preventing target windows from crossing the annual boundary.

The StableHGB grid contains 108 policy candidates:

$$3 \text{ entry ranks} \times 3 \text{ exit ranks} \times 3 \text{ confirmation lengths} \times 2 \text{ guard exposures} \times 2 \text{ smoothing windows}. \tag{7}$$

For candidate k, the cross-fold score is

$$S_k = \overline{R}_k - 0.25\,\sigma(R_k), \tag{8}$$

where R_k denotes validation cumulative return. Candidates within 0.02 of the best S_k are retained. Among them, the candidate with the least severe worst-fold drawdown, equivalently the worst-fold drawdown closest to zero, is preferred. Remaining ties are resolved by validation score, mean validation return, an auxiliary return–Sharpe score, and mean Sharpe ratio.

After policy selection, the final classifier is trained on 1,599 labeled observations whose target windows end before January 1, 2024. The 2024 labels are not incorporated into this final fit because 2024 serves as a policy-validation fold.

3.7 Backtest accounting and metrics

Let $R_t = C_t/C_{t-1} - 1$ be the close-to-close index return. With one-row execution lag, the executed exposure is

$$\widetilde{z}_t = z_{t-1}. \tag{9}$$

For a one-way proportional cost c, the strategy return is

$$R_t^{\text{strat}} = \widetilde{z}_t R_t - c\,|\widetilde{z}_t - \widetilde{z}_{t-1}|\,, \tag{10}$$

and equity evolves according to

$$V_t = V_{t-1}\left(1 + R_t^{\text{strat}}\right). \tag{11}$$

Headline results use $c = 0$. Cost sensitivity uses 5, 10, and 20 basis points. The accounting convention applies the target position recorded at row $t-1$ to the close-to-close return recorded at row t; it does not simulate a next-open fill. The test-period account also inherits the position state formed before the evaluation boundary.

Cumulative return, annualized return, maximum drawdown, and the annualized Sharpe ratio [12] are computed as

$$R_{\text{cum}} = \frac{V_T}{V_0} - 1, \tag{12}$$

$$R_{\text{ann}} = \left(\frac{V_T}{V_0}\right)^{252/N} - 1, \tag{13}$$

$$\text{MDD} = \min_t \left(\frac{V_t}{\max_{s\le t} V_s} - 1\right), \tag{14}$$

$$\text{Sharpe} = \sqrt{252}\,\frac{\overline{R^{\text{strat}}}}{s(R^{\text{strat}})}. \tag{15}$$

The Sharpe calculation assumes a zero risk-free rate.

4 Experiment

4.1 Experimental setup

Table 3 summarizes the chronological protocol. The labeled predictive test contains 316 observations from January 2, 2025 to April 24, 2026, whose target windows end by May 6, 2026. The investment backtest contains 321 trading observations through May 6, 2026. Initial capital is 100,000 CNY; this scale affects equity levels but not percentage metrics.

Table 3: Chronological experimental protocol.

Item	Setting
Raw daily data	2017-02-03 to 2026-05-06; 2,246 records
Feature-complete data	2017-06-01 to 2026-05-06
Final model training	1,599 labels with target end before 2024-01-01
Rolling policy validation	Non-overlapping annual windows for 2022, 2023, and 2024
Labeled predictive test	2025-01-02 to 2026-04-24; 316 labels
Investment backtest	2025-01-02 to 2026-05-06; 321 observations
Initial capital	100,000 CNY
Headline transaction cost	0 basis points
Execution convention	One-row-lagged close-to-close accounting

Panel A contains four traditional timing baselines: buy-and-hold, a 20-day moving-average rule, a 20-day momentum rule, and their conjunction. Panel B contains logistic regression, random forest, gradient boosting, histogram gradient boosting, and LightGBM. All five models use the same 20 input features. For each model, 10,656 candidates are evaluated across five standard mapping families: linear clipped, rank linear, sigmoid, power, and threshold. Across five models and three folds, Panel B therefore contains 159,840 candidate–fold validation records. The Relative Signal Stabilizer and Trend Position Guard are excluded from Panel B.

4.2 Baseline evaluation

Table 4 consolidates the traditional and machine-learning baselines into one comparison. Buy-and-hold has the highest cumulative return among the traditional rules, while MA20 timing records the least severe drawdown in Panel A. Among the validation-selected machine-learning baselines, gradient boosting has the highest cumulative return, histogram gradient boosting has the least severe drawdown, and LightGBM has the highest test ROC-AUC.

Table 4: Traditional and validation-selected machine-learning baselines. A dash denotes a metric that is not applicable to a rule-based strategy.

Strategy	Mapping or rule	Valid score	Cum. return	Ann. return	Max DD	Sharpe	AUC
Panel A: traditional timing rules							
Buy-and-hold	Constant full exposure	–	109.21%	78.52%	-17.26%	2.30	–
MA20 timing	Close above MA20	–	77.64%	57.00%	-10.90%	2.76	–
20-day momentum	Positive 20-day return	–	55.73%	41.59%	-17.91%	1.77	–
MA20 + momentum	Conjunction	–	66.94%	49.53%	-11.69%	2.38	–
Panel B: validation-selected machine-learning policies							
Logistic regression	Power	0.2297	98.57%	71.35%	-14.62%	2.31	0.623
Random forest	Rank linear	0.2077	85.39%	62.36%	-9.43%	2.82	0.645
Gradient boosting	Linear clipped	0.2575	105.29%	75.88%	-12.21%	2.72	0.625
Histogram GB	Threshold	0.1528	90.38%	65.78%	-9.29%	2.81	0.643
LightGBM	Rank linear	0.2010	88.72%	64.64%	-10.48%	2.81	0.652

Figure 1 groups the three baseline views that were previously separated across multiple pages. Panels (a) and (b) show the corresponding equity paths, while panel (c) places every final reported strategy in the same risk–return plane. This arrangement allows the numerical comparison in Table 4 to be read together with the temporal and cross-strategy evidence.

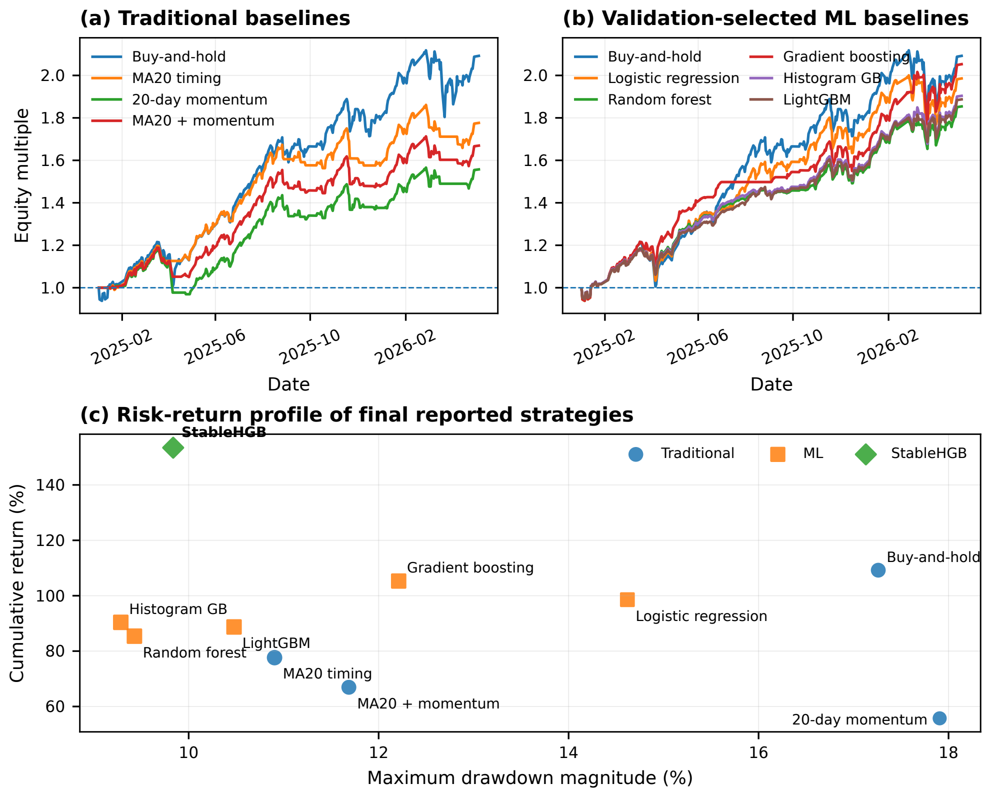

Figure 1: Baseline evaluation. (a) Equity curves for the traditional timing rules. (b) Equity curves for the validation-selected machine-learning baselines. (c) Risk–return profile of all final reported strategies, where marker size reflects the Sharpe ratio.

4.3 Main results and predictive diagnostics

Table 5 places the principal trading comparison and the predictive diagnostics in a single result block. Under the zero-cost accounting convention, StableHGB records 153.44% cumulative return, 107.52% annualized return, a -9.84% maximum drawdown, and a 3.38 Sharpe ratio. Its cumulative return exceeds buy-and-hold by 44.22 percentage points and the validation-selected gradient boosting baseline by 48.15 percentage points.

The predictive diagnostics qualify this result. The positive-label rate in the held-out labeled test set is 70.89%, so an always-positive classifier obtains 70.89% accuracy. StableHGB obtains 63.92% accuracy at a fixed 0.5 threshold but an ROC-AUC of 0.635. The model is therefore used primarily as a ranking signal; the trading result is not evidence of superior fixed-threshold classification accuracy.

Table 5: Main test-period trading comparison and predictive diagnostics. Excess return is measured in percentage points relative to buy-and-hold.

(a) Trading performance

Strategy	Cum. return	Ann. return	Max DD	Sharpe	Excess
Buy-and-hold	109.21%	78.52%	-17.26%	2.30	0.00 pp
Validation-selected GB	105.29%	75.88%	-12.21%	2.72	-3.92 pp
StableHGB	**153.44%**	**107.52%**	-9.84%	**3.38**	**+44.22 pp**

(b) Predictive diagnostics

Metric	Value	Metric	Value
Positive-label rate	70.89%	Majority-class accuracy	70.89%
StableHGB accuracy	63.92%	StableHGB ROC-AUC	0.635

Figure 2 groups the three views required to interpret the principal result. Panel (a) compares terminal wealth, panel (b) compares drawdown paths, and panel (c) aligns the executed exposure with StableHGB equity. StableHGB finishes with the highest terminal wealth, but the equity curves overlap during parts of the earlier period; the comparison therefore concerns terminal performance rather than continuous dominance.

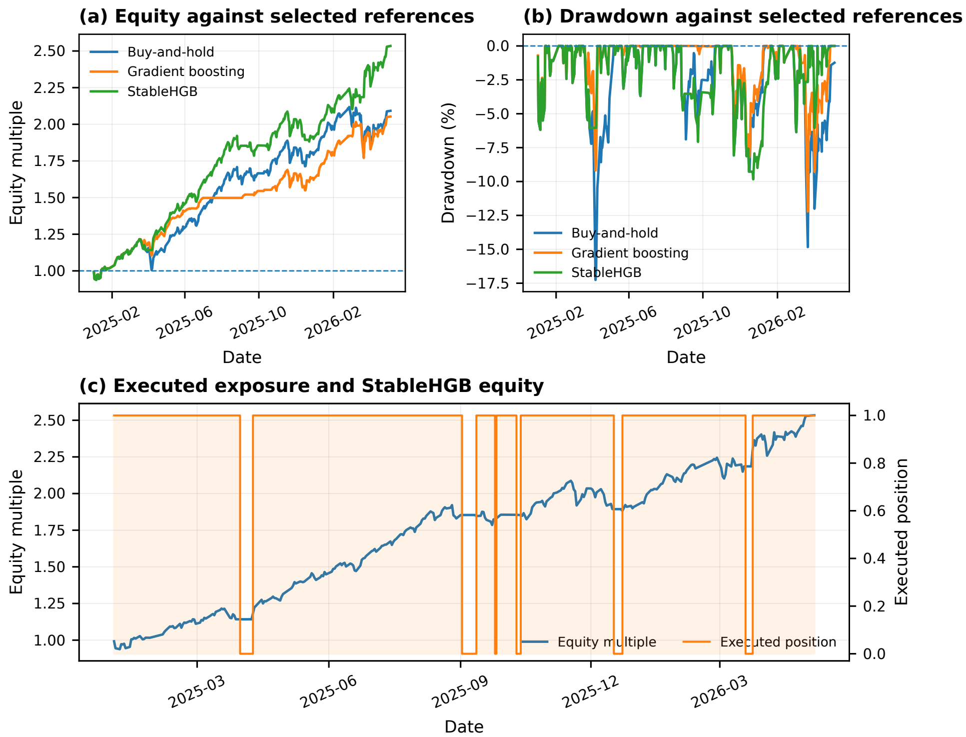

Figure 2: StableHGB test-period behavior. (a) Equity against buy-and-hold and the validation-selected gradient boosting reference. (b) Drawdown against the same references. (c) Executed binary exposure and StableHGB equity after the configured one-row lag.

4.4 Validation, ablation, and transaction-cost sensitivity

The three robustness views are reported together in Table 6. The selected policy produces positive cumulative return in every validation fold, but it is not uniformly superior to buy-and-hold. In 2023, StableHGB returns 26.80% while buy-and-hold returns 47.60%; in 2024, the selected policy reaches a maximum drawdown of -40.84%. The final-period drawdown should therefore not be interpreted as a stable risk ceiling.

The ablation rows are diagnostic rather than causal because the configurations were not selected under a fully symmetric factorial design. Relative to the standard rank-linear policy, the RSS-only configuration increases cumulative return from 82.20% to 121.51% and reduces maximum drawdown from -10.48% to -7.52%. Adding only the trend guard raises cumulative return to 106.66% but deepens maximum drawdown to -12.58%. The full policy has the highest cumulative return among the evaluated configurations.

The cost panel applies a simplified proportional charge to absolute changes in executed exposure. At 20 basis points one way, cumulative return declines by 5.98 percentage points and maximum drawdown deepens by 0.36 percentage points relative to the zero-cost result. This calculation does not model bid–ask spread, slippage, market impact, cash yield, or tracking-instrument costs.

Table 6: Rolling-validation behavior, transaction-cost sensitivity, and component ablation. The return–Sharpe score is an auxiliary fold-level tie-break statistic.

(a) Rolling-validation folds

Fold	Cum. return	Return–Sharpe	Max DD	Sharpe	B&H return
2022	30.99%	1.36	-22.29%	1.18	22.20%
2023	26.80%	0.70	-7.39%	1.99	47.60%
2024	43.76%	3.56	-40.84%	1.06	9.33%

(b) One-way cost sensitivity

Cost	Cum. return	Max DD
0 bps	153.44%	-9.84%
5 bps	151.93%	-9.93%
10 bps	150.43%	-10.02%
20 bps	147.46%	-10.20%

(c) Component ablation at zero transaction cost

Configuration	Position rule	Valid score	Cum. return	Excess	Max DD	Sharpe
Buy-and-hold	Constant full	–	109.21%	0.00 pp	-17.26%	2.30
Validation-selected GB	Linear clipped	0.2575	105.29%	-3.92 pp	-12.21%	2.72
Stable classifier + standard	Rank linear	0.1993	82.20%	-27.01 pp	-10.48%	2.65
Standard + guard	Rank linear + guard	0.2237	106.66%	-2.55 pp	-12.58%	2.67
RSS only	Relative stabilizer	0.2072	121.51%	$+12.30$ pp	-7.52%	3.59
StableHGB full	**RSS + guard**	**0.3164**	**153.44%**	$+44.22$ **pp**	-9.84%	**3.38**

4.5 Discussion and limitations

The results support three observations. First, the classifier's fixed-threshold accuracy is not the source of the trading advantage; the useful signal is better characterized by probability ranking and by the policy selected on rolling validation. Second, the RSS-only ablation provides the strongest observed Sharpe ratio and the least severe drawdown among the learned configurations, whereas the trend guard primarily improves upside participation and can increase drawdown. Third, the full policy records the highest test-period terminal wealth among the evaluated methods, but its validation behavior varies substantially across years.

Several limitations remain. The final model is trained on only 1,599 labels and is evaluated on a single index. The 108 policy candidates and the much larger Panel B policy grid create residual selection-overfitting risk despite rolling validation. The 2024 validation drawdown shows that the low final-period drawdown is not stable across all observed regimes. The one-row-lagged close-to-close convention does not simulate a next-open transaction after observing the previous close, and the test account inherits a pre-test position state. The cost model excludes spread, slippage, market impact, financing, cash yield, and tracking-instrument constraints. The final classifier is not refitted with 2024 labels after policy selection. Probability calibration and statistical significance are not evaluated. Finally, the held-out period is one favorable historical realization; the reported differences should not be read as guarantees of future performance.

4.6 Reproducibility

The project provides deterministic scripts, a pinned software environment, recorded data and output hashes, and a reproduction checker. The principal commands are:

```
python scripts/reproduce_stable_hgb.py
python scripts/run_all_experiments.py -workers 10
```

The reproduction checker verifies selected data and output files against recorded SHA256 hashes. Reproducibility remains conditional on the pinned Python and package environment because tree-ensemble implementations can exhibit version- and platform-dependent numerical differences.

5 Conclusion

This paper presents StableHGB, a complete index-timing pipeline that combines a histogram gradient boosting direction classifier with two explicit position-policy components. The Relative Signal Stabilizer converts probability estimates into a prior-relative rank, applies consecutive confirmation, and maintains an entry–exit deadband. The Trend Position Guard restores full exposure under bullish moving-average alignment. Policy parameters are selected through annual rolling validation rather than on the final investment period.

Under the reported zero-cost close-to-close accounting convention, StableHGB reaches 153.44% cumulative return, a 3.38 Sharpe ratio, and a -9.84% maximum drawdown over the held-out investment period, compared with 109.21%, 2.30, and -17.26% for buy-and-hold. The RSS-only diagnostic has the highest observed Sharpe ratio, while the full policy has the highest terminal wealth. Cost sensitivity remains moderate under the simplified proportional-turnover model.

The results demonstrate the value of treating probability-to-position mapping as a separately validated component rather than using a fixed classification threshold. At the same time, the 2024 validation drawdown, below-majority fixed-threshold accuracy, simplified execution assumptions, and single-index evaluation limit the strength of the conclusions. Future work should evaluate next-open or intraday execution, nested policy selection, probability calibration, refitting with the latest pre-test data, multiple indices and market regimes, and uncertainty estimates for performance differences.

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